Engine Condition Trend Monitoring/Flight Operations Quality Assurance Data Capture (S.O.O. 6288) – General Data

1. General

The Engine Condition Trend Monitoring (ECTM)/Flight Operations Quality Assurance (FOQA) Data Capture (S.O.O. 6288) system consists of an IONode recording unit and a multi element GPS/WiFi antenna. The IONode continuously records data from the aircraft Flight Data Recorder, and its own internal sensors. Data is subsequently downloaded to a Personal Computer (PC) for analysis.

2. Description

The ECTM/FOQA Data Capture system is a data storing and forwarding system. The IONode unit provides multiple inputs (digital and analog) for gathering data from onboard sensory equipment. The IONode is designed to do local logging to internal, on-board storage or an optional USB flash drive. In this installation the IONode records FDR Data, GPS position data and data from its internal accelerators.

The system consists of two major components; the IONode unit and combined GPS/Iridium/Wi-Fi Antenna The IONode unit is located on the bottom surface of the upper, left hand avionics shelf in the forward cargo compartment. The combined GPS/Iridium/Wi-Fi antenna is located on the top surface of the fuselage at station 246.50 near the right hand side of the aircraft. The ECTM/FOQA circuit breaker (Q4), located on the avionics circuit breaker panel is powered through CB230.

The ECTM/FOQA Data Capture system is a standalone installation that interfaces with one other aircraft system. The IONode unit receives all of its information from the Flight Data Recorder, interfacing with the ARNIC 717 output port on the Flight Data Recorder.

3. Equipment and Materials

- Cleaner MIL–C–25769
- Isopropanol
- Multimeter, 1 megaohm range
- Personal Computer (PC) with a Wi-Fi adapter, internet connection and a web browser
- Latitude Technologies registered account

4. Cleaning

The antenna may be cleaned only with the approved cleaning products listed in Para 3. Refer to 12-30-01, Aircraft External Cleaning – General Data, for cleaning procedures. Limit cleaning to wiping with moistened cloths.

5. Adjustment/Test

A. Function Test of the ECTM/FOQA Data Capture System

1) General

a) Switch aircraft power ON.

b) Open the ECTM/FOQA (Q4) circuit breaker on the avionics circuit breaker panel and ensure that all the LEDs on the IONode are OFF.

c) Close the ECTM/FOQA (Q4) circuit breaker on the avionics circuit breaker panel and ensure that there is activity on the LED's on the IONode (LED L1 reports Wi-Fi status, LED L2 reports GPS status, LED L3 reports overall system status).

2) No Engine Test

Note This test requires availability of the Ground Support Computer. Communications between the aircraft and the person monitoring the Ground Support Computer are required. The computer software and the IONode must be configured prior to this test.

a) Switch the aircraft power ON.

b) Establish communication between the aircraft and the person at the Ground Support Computer. Link the IONode either by Wi-Fi or USB-B connection.

c) Check that the green L2 and L3 status lights on the IONode are ON and steady.

d) When using Wi-Fi to connect to the Ground Support Software, enable IONode Wi-Fi by holding the "ECTM/FOQA TRANSMIT SWITCH" ON for ten seconds.

e) Check that the green L1 status light on the IONode is ON steady indicating that a Wi-Fi connection is established.

f) Check that the Ground Support Computer is receiving data. Verify that the position, speed (0), and altitude are correct for the aircraft's current position.

g) Switch OFF aircraft electrical system power. (Refer to 24-00-00, Electrical Power – General Data).

6. Removal/Installation

A. Combined GPS/Iridium/Wi-Fi Antenna

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the ECTM/FOQA (Q4) circuit breaker on the avionics circuit breaker panel and fit safety clip. If S.O.O. 6298 is installed, open the SAT COM (N4) circuit breaker and fit safety clip.

c) Remove the four (4) screws securing the antenna to the airframe.

d) Carefully cut the sealant from around the base of the antenna.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connections.

f) Disconnect the coaxial connectors.

Note Stow connectors as required.

g) Remove the antenna.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the ECTM/FOQA (Q4) circuit breaker is opened and fitted with safety clip. If S.O.O. 6298 is installed, ensure the SAT COM (N4) circuit breaker is opened and fitted with safety clip.

c) Remove non-conductive material (e.g. anodic films, primer, paint, etc) from the electrical bond contact area in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data. Then apply chemical conversion coating (C10) in accordance with 20-30-02, Chemical Conversion Coatings (Alodine) – Manual Application, within 8 hours of stripping high resistance films.

d) Solvent clean the electrical contact bond areas immediately before installation. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data).

e) Move the antenna close to the fuselage and connect the coaxial connectors.

f) Secure the antenna to the fuselage using the four (4) mounting screws.

g) Solvent clean the exposed areas of the airframe around the antenna. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data).

h) Touch up exposed areas of the airframe around the antenna, if required, with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

i) If the affected area requires paint touch up, brush coat the area with the top coat paint system matching the existing.

j) Fillet seal around the base of the antenna with MIL–PRF–81733 sealant in accordance with 20-50-01, Weather Sealing – General Data. The aircraft will be ready to fly 24 hours after application of the sealant.

k) Do a resistance measurement between the antenna and the airframe. Resistance must be less than 3 milliohms in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data.

l) Remove the safety clip and close the ECTM/FOQA (Q4) circuit breaker. If S.O.O. 6298 is installed, remove safety clip and close the SAT COM (N4) circuit breaker.

m) Do a function test of the ECTM/FOQA Data Capture system per Para 5.A.1).

B. IONode Unit

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the ECTM/FOQA (Q4) circuit breaker on the avionics circuit breaker panel and fit safety clip. If S.O.O. 6298 is installed, open the SAT COM (N4) circuit breaker and fit safety clip.

c) Disconnect the connectors.

Note Stow connectors as required.

d) Remove the four (4) screws and washers securing the IONode in position on the bottom face of the upper left avionics shelf in the FWD cargo compartment.

e) Carefully remove the IONode unit.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the ECTM/FOQA (Q4) circuit breaker is opened and fitted with safety clip. If S.O.O. 6298 is installed, ensure the SAT COM (N4) circuit breaker is opened and fitted with safety clip.

c) Carefully position the IONode over the four (4) potted mounting holes on the bottom face of the upper left avionics shelf in the FWD cargo compartment.

d) Install the four (4) screws and washers to secure the IONode in position.

e) Connect the connectors.

f) Remove the safety clip and close the ECTM/FOQA (Q4) circuit breaker. If S.O.O. 6298 is installed, remove safety clip and close the SAT COM (N4) circuit breaker.

Note Calibrate in accordance with IONode Installation and Operational Manual #740–0193 Revision 1.16 or later.

g) Do a function test of the ECTM/FOQA system as per Para 5.A.1).

7. Troubleshooting

A. IONode Unit

1) If the IONode is deemed inoperative check the following:

a) Ensure that the ECTM/FOQA (Q4) circuit breaker is closed.

b) Check that the IONode is properly secured and grounded to the airframe.

c) Ensure that all connectors are properly mated and tightened.

d) Visually inspect the IONode for damaged wires or contacts and ensure no short circuits have formed.

e) Check that the Combined GPS/Iridium/Wi-Fi Antenna is properly secured and bonded to the airframe.

f) Visually inspect the IONode and antenna for damaged wires or contacts.

g) If troubleshooting at the wire level is required, refer to the applicable diagrams in the Wiring Diagrams Manual, PSM 1-64-2W.

h) Do a function test of the ECTM/FOQA system as per Para 5.A.1).

i) If repair or service is required, send the IONode to appropriately rated avionics repair facility.

2) The LED indicators on the IONode may assist in problem diagnosis:

LED L1	**Wi-Fi STATUS**
LED Off Solid	Wi-Fi subsystem disabled.
LED On Solid	Wi-Fi subsystem enabled and connected to Base Station Software (socket open).
LED Blinking once per second	Wi-Fi subsystem enabled, or searching for access point, or attempting to open socket.
LED Flickering	Wi-Fi subsystem enabled and transferring data.

LED L2	**GPS STATUS**
LED On Solid	GPS fix acquired.
LED Off Solid	No GPS source, or GPS using external source and cannot parse GPS string.
LED Blinking once per two seconds	GPS present and fix not acquired.

LED L3	**SYSTEM STATUS**
LED On Solid	System running normally.
LED Off Solid	System Off.

Table 1

Symptom	Diagnosis
IONODE L3 Light won't turn ON.	Verify correct power and ground wiring.
IONODE L2 LED Light indicates "No GPS Fix" blink pattern.	Make sure that the triple Iridium/GPS/Wi-Fi antenna is properly connected to the IONODE and that the antenna has a clear view of the sky.
IONODE L1 LED will not turn ON.	Make sure that the correct Configuration Settings file has been installed on the IONODE and that the associated Wi-Fi enable signals are present.
IONODE L1 LED blinking even though there is an access point nearby.	Make sure that the correct Configuration Settings file has been installed on the IONODE. Double check the SSID, Password and IP settings that were sent to Latitude Technologies Support to create the Configuration Settings. Make sure that the triple Iridium/GPS/Wi-Fi antenna is properly connected to the IONODE and that the antenna is located in an area with good exposure to the access point signal (this can be verified in the location of the Wi-Fi antenna using a smart phone or other Wi-Fi enabled mobile device).
IONODE L2 LED won't turn ON when SkyNode is installed in the same aircraft.	When S.O.O. 6288 and S.O.O. 6298 are installed in the same aircraft, the GPS position for the IONode is obtained from the S200 SkyNode. Verify GPS data wiring between the SkyNode and IONode.
IONODE log files do not show sensor readings.	If operating on the ground, make sure to place the IONode in Maintenance Mode (this can be done through the Base Station Software).
IONODE IMU yaw/pitch/roll readings are swapped or appear to be incorrect.	Make sure that the IONode orientation procedure has been completed, that the recorded values were passed to a Latitude Technical Support Representative, and that the returned Configuration Settings files has been installed in the IONode. If the values are still incorrect, an additional 2D calibration procedure is required. Contact Latitude Technologies Technical Support for calibration assistance. (Refer to 23-15-00 – Satellite Communication System (S.O.O. 6298) – General Data for full contact information).

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